

Problem

Obtain the sinusoids corresponding to each of the following phasors:

(a) $\mathbf{V}_1 = 60 \angle 15^\circ \text{ V}, \omega = 1$

(b) $\mathbf{V}_2 = 6 + j8 \text{ V}, \omega = 40$

(c) $\mathbf{I}_1 = 2.8e^{-j\pi/3} \text{ A}, \omega = 377$

(d) $\mathbf{I}_2 = -0.5 - j1.2 \text{ A}, \omega = 103$

Step-by-step solution

Step 1 of 4

(a)

$$\mathbf{V}_1 = 60 \angle 15^\circ \text{ V}, \omega = 1$$

The corresponding sinusoidal is,

$$v_1(t) = 60 \cos(t + 15^\circ) \text{ V}$$

Step 2 of 4

(b)

$$\mathbf{V}_2 = 6 + j8, \omega = 40$$

$$V = 10 \angle 53.13^\circ, \omega = 40$$

The corresponding Sinusoid is,

$$v_2(t) = 10 \cos(40t + 53.13^\circ) \text{ V}$$

Step 3 of 4

(c)

$$\mathbf{I}_1 = 2.8e^{-j\frac{\pi}{3}}, \omega = 377$$

$$\mathbf{I}_1 = 2.8 \angle \frac{-\pi}{3}$$

$$\mathbf{I}_1 = 2.8 \angle -60^\circ$$

The corresponding sinusoid is,

$$i_1(t) = 2.8 \cos(377t - 60^\circ) \text{ A}$$

Step 4 of 4

(d)

$$I_2 = -0.5 - j1.2 \text{ A}, \quad \omega = 10^3$$

$$I_2 = 1.3 \angle -112.62^\circ \text{ A}$$

$$I_2 = 1.3 \angle 247.38^\circ \text{ A}$$

The corresponding sinusoid is,

$$I_2(t) = 1.3 \cos(10^3 t + 247.38^\circ) \text{ A}$$